

EXPERIMENT NO. : 7 (ASSIGNMENT)

DATE: 01/10/2026

1. AIM

To design a CMOS inverter repeater using minimum channel length and minimum width in 180nm technology and determine

- 1] R_{out} , (output resistance of the inverter)
- 2] C_{out} , (output capacitance of the inverter)
- 3] intrinsic delay t_p , (intrinsic delay of the inverter)
- 4] R_n , R_p (resistance of NMOS and PMOS)
- 5] k (optimum number of repeater stages)
- 6] optimal repeater size for minimum propagation delay
- 7] propagation delay of the cascaded driver

to calculate and compare the propagation delay without repeater insertion, with optimally sized repeater insertion, and using a cascaded driver.

2. TOOL

Cadence Virtuoso

3. THEORY

1. Unbuffered Interconnect Delay: A long interconnect exhibits distributed resistance R_{int} and capacitance C_{int} , causing delay to scale quadratically with wire length. Driven by a minimum-sized inverter (R_{out} , C_{out}), the total unbuffered delay is given by
$$T_{50\%} = 0.7R_{out}(C_{int} + C_{out}) + R_{int}(0.4C_{int} + 0.7C_{out}).$$

2. Optimal Repeater Insertion: Inserting intermediate inverters divides the line into k shorter segments, converting quadratic delay growth into a linear relationship. For k segments and sizing factor h , the total delay is $T_{50\%} = k \left[0.7 \frac{R_{out}}{h} \left(\frac{C_{int}}{k} + hC_{out} \right) + \frac{R_{int}}{k} \left(0.4 \frac{C_{int}}{k} + 0.7hC_{out} \right) \right]$. Minimizing with respect to k and h yields $k = \sqrt{\frac{0.4R_{int}C_{int}}{0.7R_{out}C_{out}}}$ and $h = \sqrt{\frac{R_{out}C_{int}}{R_{int}C_{out}}}$.

3. Cascaded Driver: A tapered chain of n progressively sized inverters (scaling factor $f = e \approx 2.7182$) drives the interconnect from the source without intermediate repeaters. The optimal number of stages is $k_{casc} = \ln \left(\frac{C_{int} + C_L}{C_{out}} \right)$ and the total delay is

$$T_{50\%} = 0.7(n-1)fR_{out}C_{out} + \left(\frac{0.7R_{out}}{f^n - 1} + 0.4R_{int} \right) C_{int} + \left(\frac{0.7R_{out}}{f^n - 1} + 0.7R_{int} \right) C_L.$$

4. TABLES

Table 4.1: Design Parameters

given parameter	value	unit
technology	180	nm
V_{dd}	1.8	V
V_{tn}	0.45	V
$ V_{tp} $	0.45	V
minimum width of nmos	400	nm
minimum width of pmos	400	nm
k_n'	70	$\mu\text{A}/\text{V}^2$
k_p'	20	$\mu\text{A}/\text{V}^2$
R_{int}	68	$\text{k}\Omega$
C_{int}	60	fF

Table 4.2: Derived and observed intrinsic parameters

derived and observed parameter	value	unit
t_{phl}	9.8587	ps
t_{plh}	27.33919	ps
t_p	18.5989	ps
R_n	4.762	$\text{k}\Omega$
R_p	16.67	$\text{k}\Omega$
R_{out}	10.716	$\text{k}\Omega$
C_n	3.0004	fF
C_p	2.3768	fF
C_{out}	3.0004	fF
k (optimal size repeater)	8	unitless
h (optimal size repeater)	3.15	unitless
k_{casc} (cascaded driver)	3	unitless
f (cascaded driver)	2.7182	unitless

Table 4.3: Result

parameter	without inv/buf insertion	optimal size repeaters	cascaded driver	unit	difference (for optimal size repeaters)	difference (for cascaded driver)
t_{phl}	2.882	1.01343	2.93671	ns	-64.8359%	+1.8983%
t_{plh}	3.119	1.07319	3.18355	ns	-65.5919%	+2.0696%
t_p	3.0005	1.0433	3.0601	ns	-65.2291%	+1.9863%

Table 4.4: comparison of theoretical and practical values

parameter	theoretical value (T_p)	practical value (t_p)	unit	signed error
without inv/buf insertion	2.2473	3.0005	ns	+33.5158%
optimal size repeaters	0.9768	1.0433	ns	+6.8079%
cascaded driver	1.8152	3.0601	ns	+68.582%

5. CALCULATION

$$\begin{aligned}
 R_n &= 1000/(k_n * (W/L) * (V_{dd} - V_{tn})) \\
 &= 1000/(70 * (400/180) * (1.8 - 0.45)) \\
 &= 4.7619048k\Omega \approx 4.762k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_p &= 1000/(k_p * (W/L) * (V_{dd} - V_{tp})) \\
 &= 1000/(20 * (400/180) * (1.8 - 0.45)) \\
 &= 16.6666667k\Omega \approx 16.67k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_{out} &= (R_n + R_p)/2 \\
 &= (4.7619048 + 16.6666667)/2 \\
 &= 10.7142858k\Omega \approx 10.714k\Omega
 \end{aligned}$$

$$\begin{aligned}
 C_{out1} &= t_{phl}/(0.69 * R_n) \\
 &= 9.8587/(0.69 * 4.7619048) \\
 &= 3.0004739fF \approx 3fF
 \end{aligned}$$

$$\begin{aligned}
 C_{out2} &= t_{plh}/(0.69 * R_p) \\
 &= 27.33919/(0.69 * 16.6666667)
 \end{aligned}$$

$$= 2.3773209fF \approx 2.3773fF$$

$$C_{out} = \max(C_{out1}, C_{out2})$$

$$= \max(3.0004739, 2.3773209)$$

$$= 3.0004739fF \approx 3fF$$

$$t_{p(min)} = (t_{phl} + t_{plh})/2$$

$$= (9.8587 + 27.33919)/2$$

$$= 18.598945ps \approx 18.5989ps$$

$$k = \sqrt{(R_{int} * C_{int} * 0.4)/(R_{out} * C_{out} * 0.7)}$$

$$= \sqrt{(68 * 60 * 0.4)/(10.7142858 * 3.0004739 * 0.7)}$$

$$= 8.5159777 \approx 8$$

$$h = 3.15$$

$$k_{casc} = \ln((C_{int} + C_L)/C_{out})$$

$$= \ln((60 + 0)/3.0004739)$$

$$= 2.9955743 \approx 3$$

$$T_{p(no_rep)(theoretical)} = (0.7 * R_{out} * (C_{int} + C_{out})$$

$$+ R_{int} * (0.4 * C_{int} + 0.7 * C_{out}))/1000$$

$$= (0.7 * 10.7142858 * (60 + 3.0004739)$$

$$+ 68 * (0.4 * 60 + 0.7 * 3.0004739))/1000$$

$$= 2.2473261ns \approx 2.2473ns$$

$$T_{p(opt)(theoretical)} = (k * (0.7 * (R_{out}/h) * (C_{int}/k + h * C_{out})$$

$$+ (R_{int}/k) * (0.4 * C_{int}/k + 0.7 * h * C_{out}))/1000$$

$$= (8 * (0.7 * (10.7142858/3.15) * (60/8 + 3.15 * 3.0004739)$$

$$+ (68/8) * (0.4 * 60/8 + 0.7 * 3.15 * 3.0004739)))/1000$$

$$= 0.9767766ns \approx 0.9768ns$$

$$T_{p(casc)(theoretical)} = (0.7 * (k_{casc} - 1) * f * R_{out} * C_{out}$$

$$+ (0.7 * R_{out}/(f * k_{casc} - 1)) + 0.4 * R_{int}) * C_{int}$$

$$\begin{aligned}
& + (0.7 * R_{out} / (f^{(k_{casc} - 1))} + 0.7 * R_{int}) * C_L) / 1000 \\
& = (0.7 * (3 - 1) * 2.7182 * 10.7142858 * 3.0004739 \\
& \quad + (0.7 * 10.7142858 / (2.7182^{(3 - 1))} + 0.4 * 68) * 60 \\
& \quad + (0.7 * 10.7142858 / (2.7182^{(3 - 1))} + 0.7 * 68) * 0) / 1000 \\
& = 1.8152429ns \approx 1.8152ns
\end{aligned}$$

$$t_{HL(no_rep)} = 2.882ns$$

$$t_{LH(no_rep)} = 3.119ns$$

$$t_{p(no_rep)} = (t_{phl_no_rep} + t_{plh_no_rep}) / 2$$

$$= (2.882 + 3.119) / 2$$

$$= 3.0005ns$$

$$t_{HL(opt)} = 1.01343ns$$

$$t_{LH(opt)} = 1.07319ns$$

$$t_{p(opt)} = (t_{phl_opt} + t_{plh_opt}) / 2$$

$$= (1.01343 + 1.07319) / 2$$

$$= 1.04331ns$$

$$t_{HL(casc)} = 2.93671ns$$

$$t_{LH(casc)} = 3.18355ns$$

$$t_{p(casc)} = (t_{phl_casc} + t_{plh_casc}) / 2$$

$$= (2.93671 + 3.18355) / 2$$

$$= 3.06013ns$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

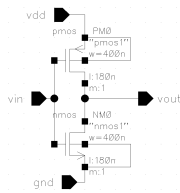


Figure 6.1: Minimum inverter circuit

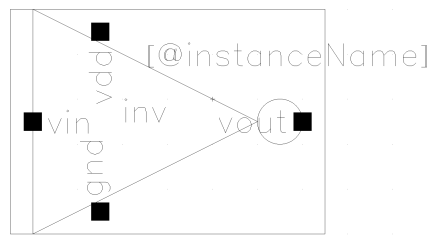


Figure 6.2: Minimum inverter symbol

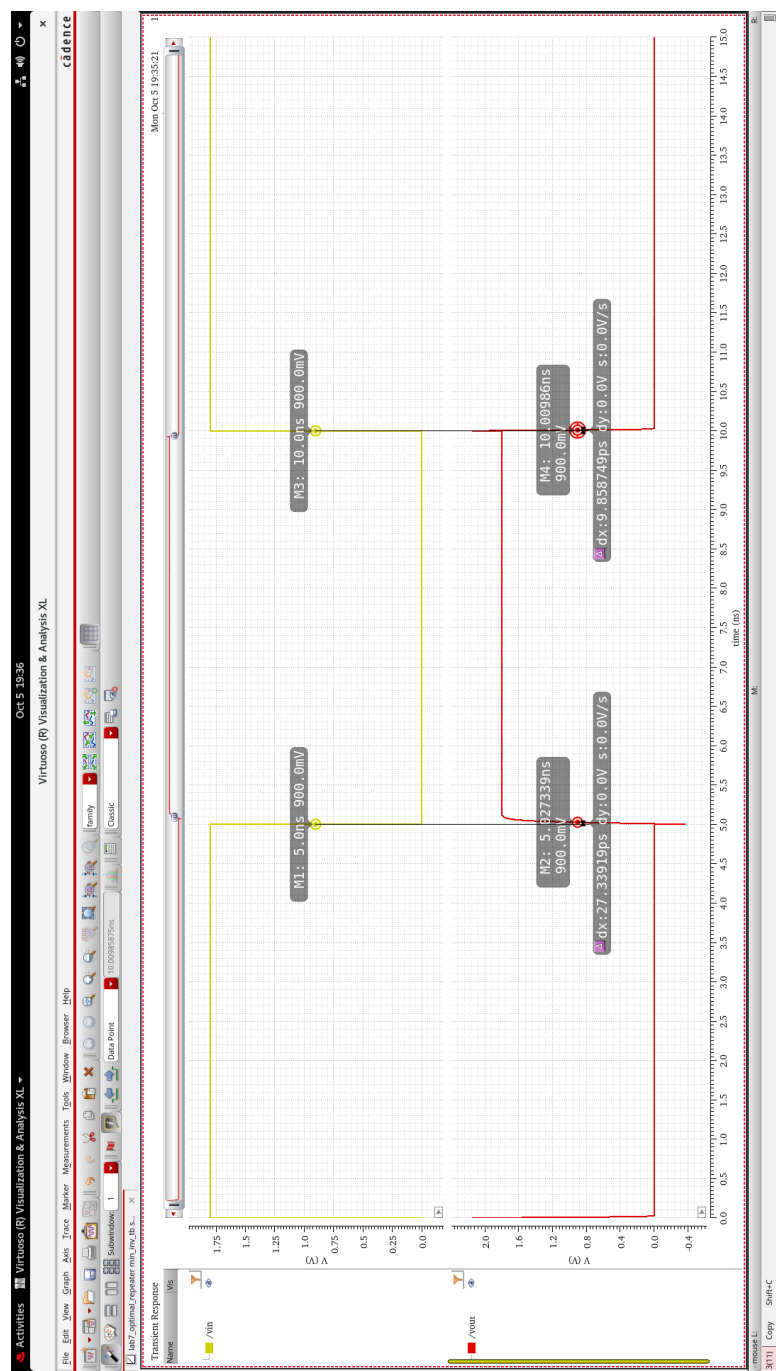


Figure 6.3: Minimum inverter waveform

$t_{PHL} = 9.8587ps, \quad t_{PLH} = 27.3391ps, \quad t_p = 18.598945ps$

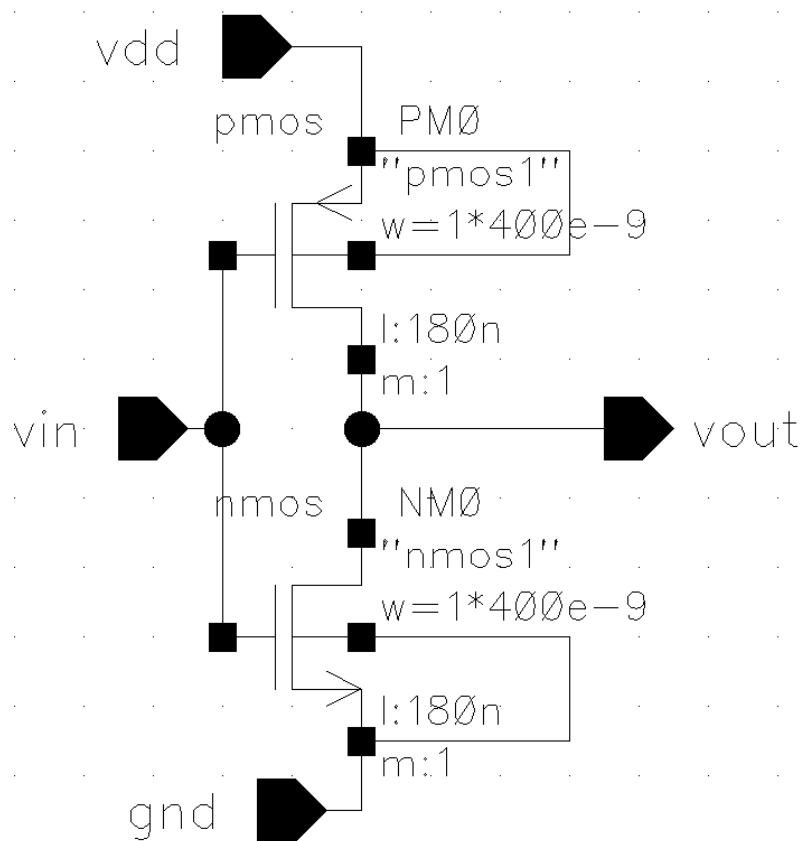


Figure 6.4: Sized Inverter circuit

cdsParam(1)

cdsParam(2)

cdsParam(3)

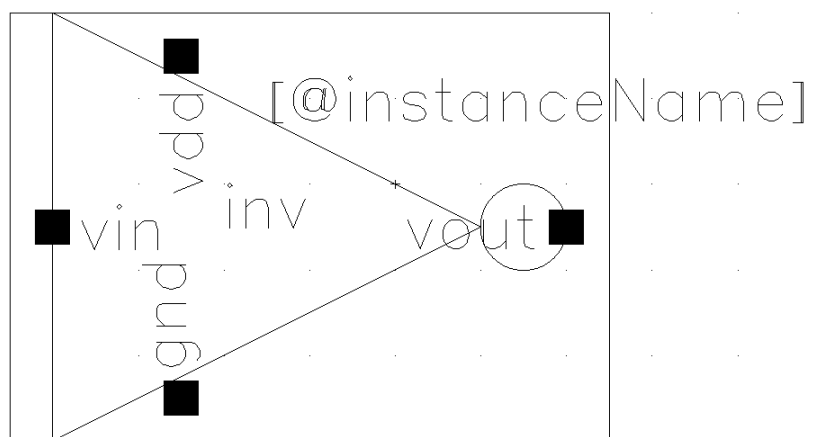


Figure 6.5: Sized Inverter Symbol

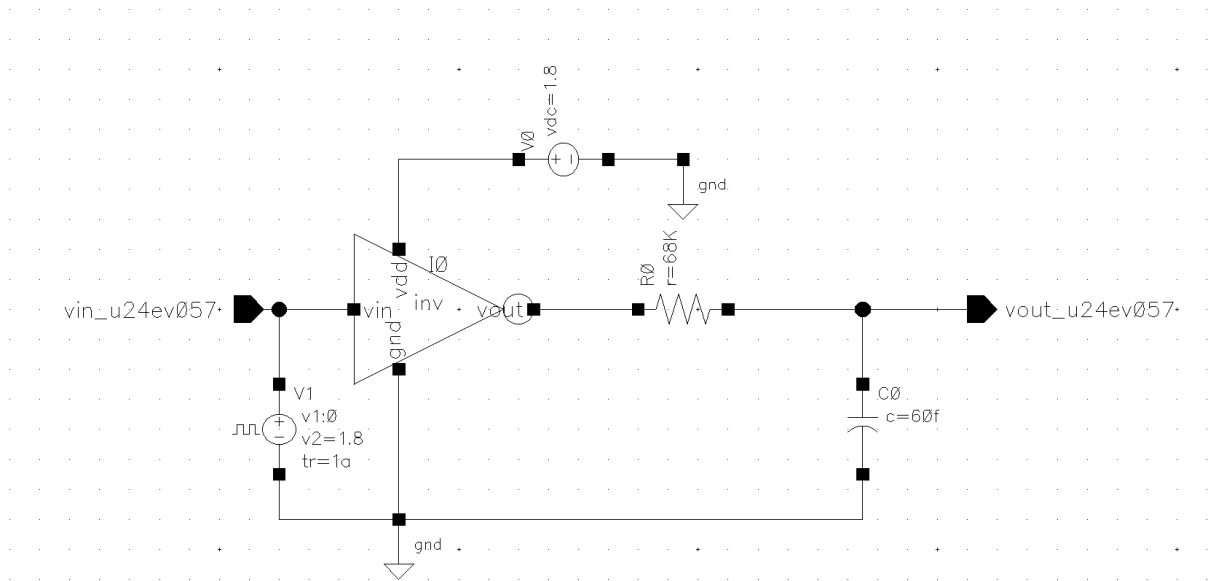


Figure 6.6: Minimum inverter with load circuit

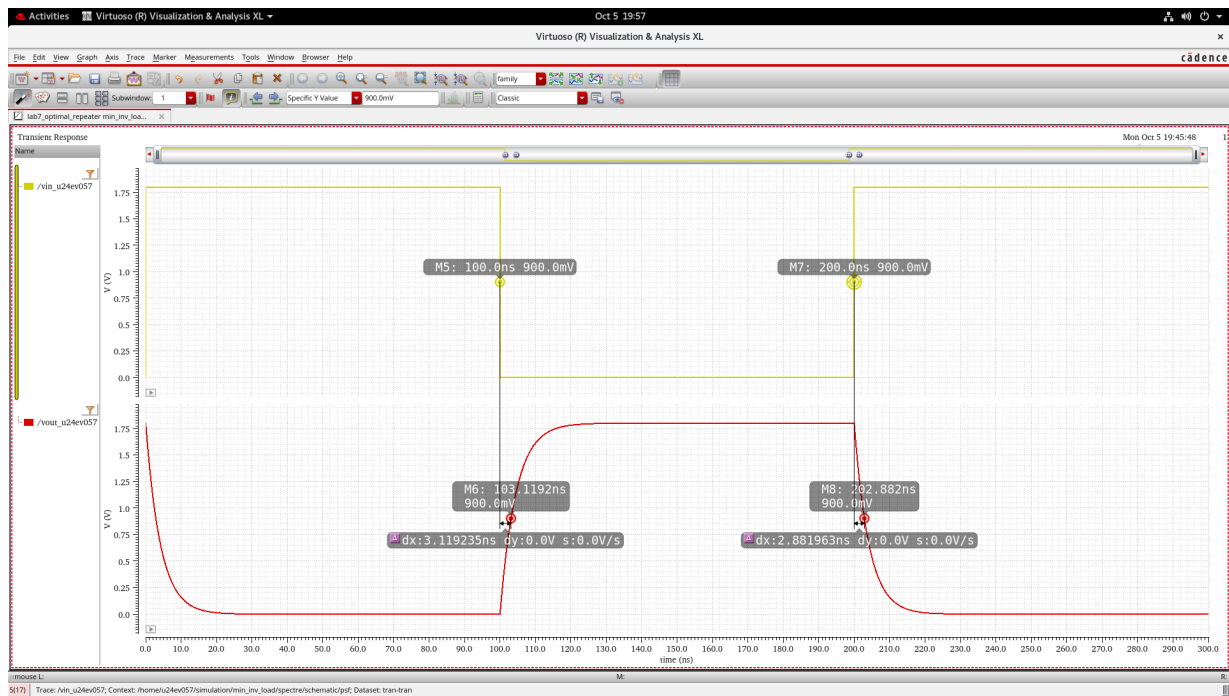


Figure 6.7: Minimum inverter with load waveform

$$t_{PHL} = 2.882\text{ns}, \quad t_{PLH} = 3.119\text{ns}, \quad t_p = 3.0005\text{ns}$$

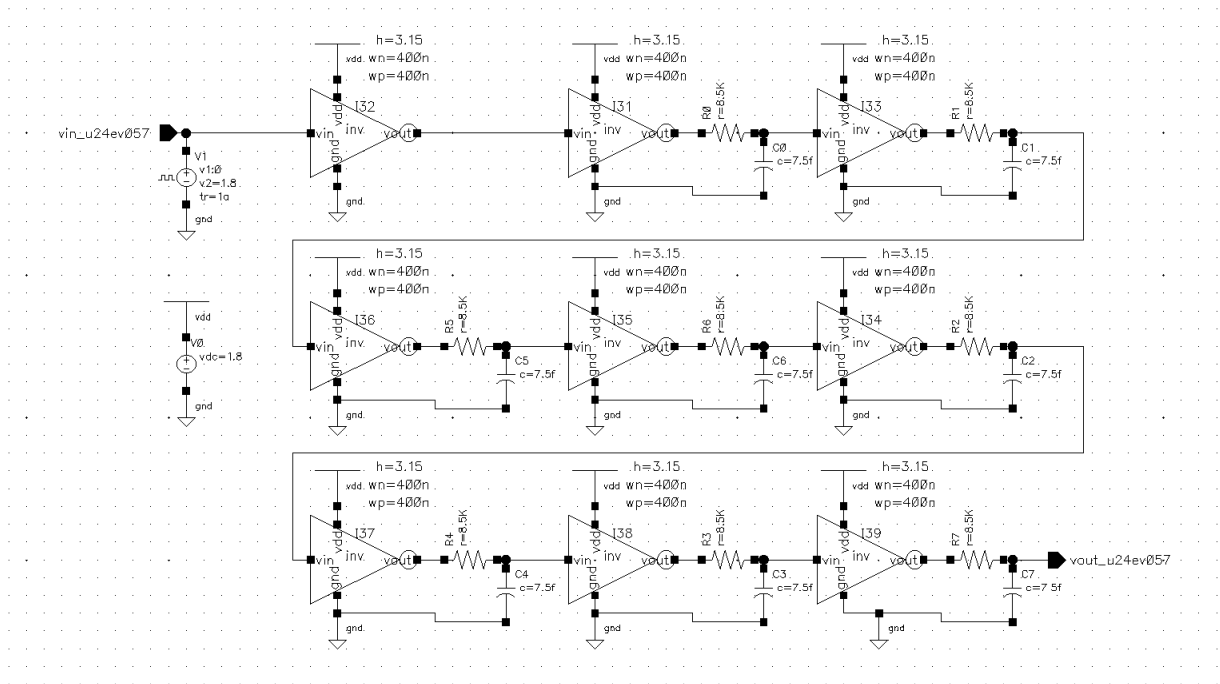


Figure 6.8: Optimal size repeater circuit

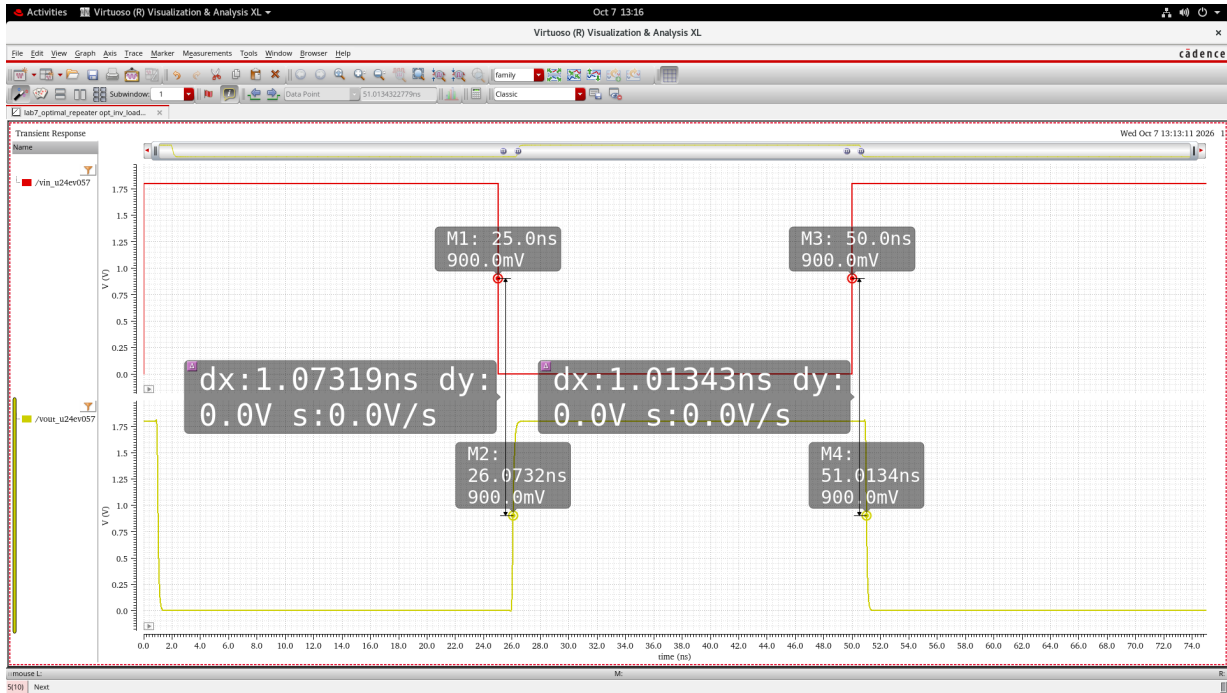


Figure 6.9: Optimal size repeater waveform

$$t_{PHL} = 1.01343\text{ns}, \quad t_{PLH} = 1.07319\text{ns}, \quad t_p = 1.04331\text{ns}$$

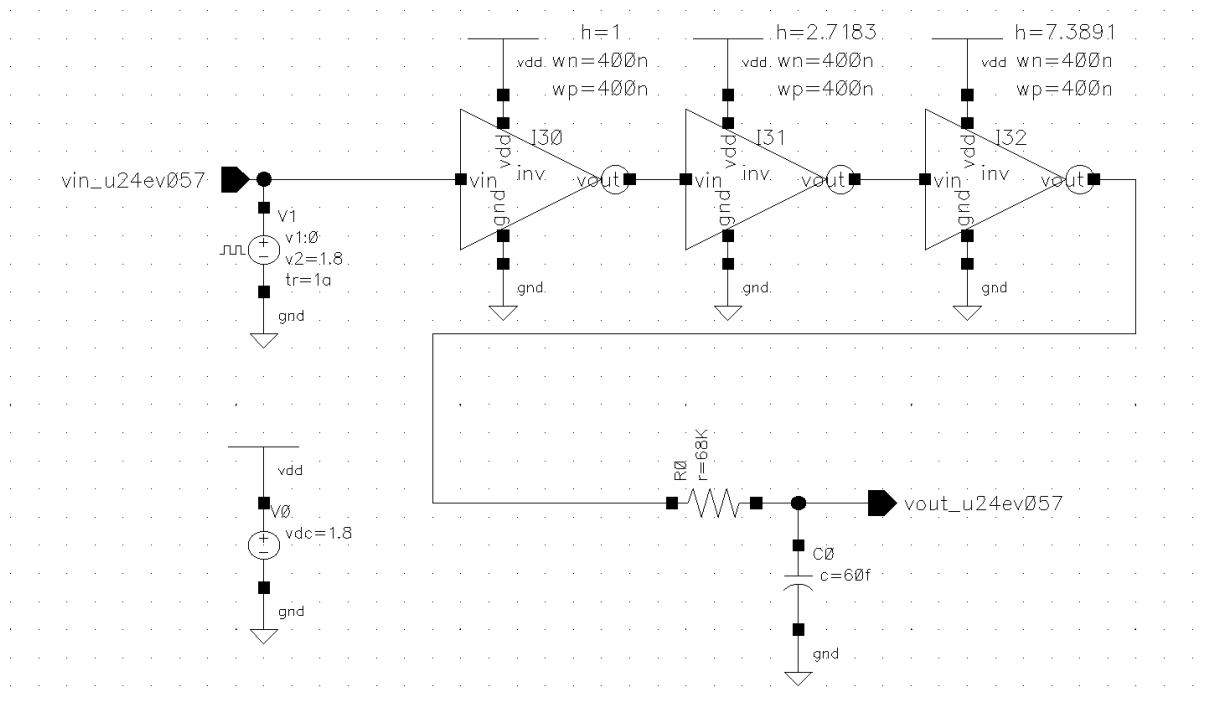


Figure 6.10: Cascaded driver circuit

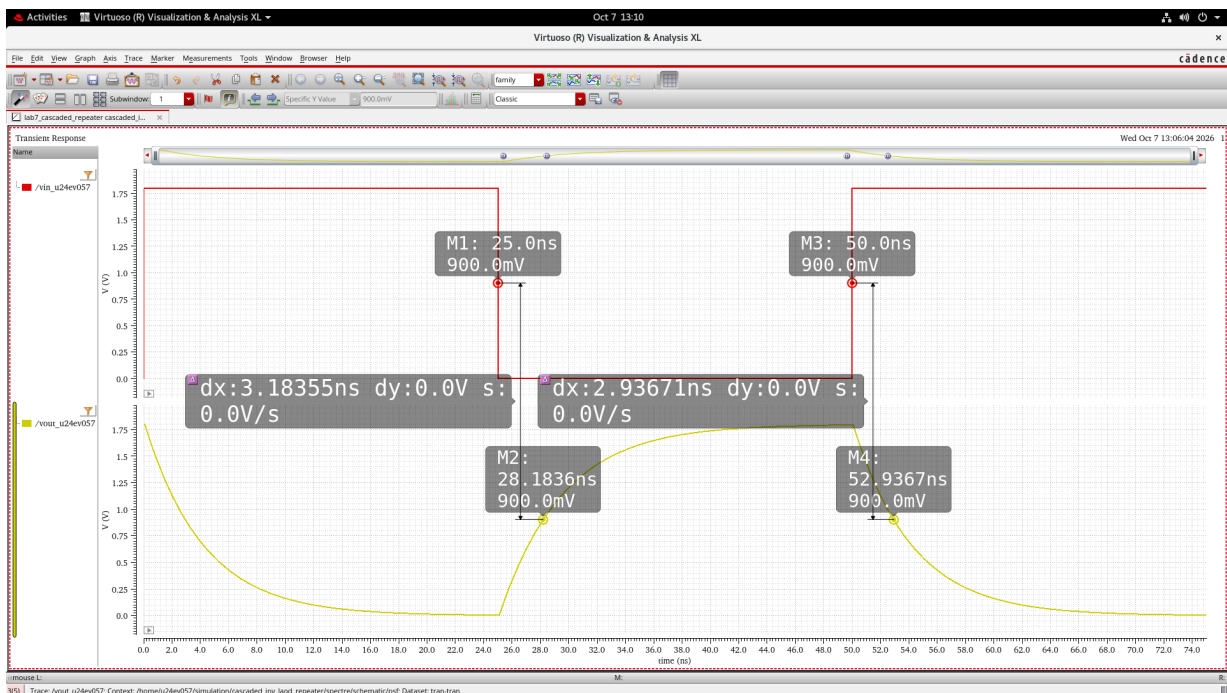


Figure 6.11: Cascaded driver Waveform

$$t_{PHL} = 2.93671\text{ns}, \quad t_{PLH} = 3.18355\text{ns}, \quad t_p = 3.06013\text{ns}$$

7. CONCLUSION

1. Optimal repeater insertion ($k = 8$, $h = 3.15$) reduced propagation delay from 3.0005ns to 1.0433ns (65.23% reduction), closely matching the theoretical delay of 0.9768ns (+6.81% error).
2. A cascaded driver ($k_{casc} = 3$) yielded 3.0601ns (theoretical 1.8152ns), proving that distributed repeaters are far superior to lumped drivers for long distributed RC interconnects.